

Testing Replication With Opensource Data

Bradley Allen

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Abstract

This project leveraged open-source data and open-source-software for replication research in psychological science. Findings included:

- Confirmatory Factor Analysis of the 21-item Depression Anxiety Stress Scale (DASS-21)
- Gender differences in personality traits
- Gender differences in experiencing depression, anxiety, and stress
- Personality as a predictor of stress for males and females



Introduction & Motivation

Problem & question: In 2011, high rates of replication failure plunged psychological research into a crisis of public trust. This project aimed to repair that trust by testing replication using open-source data and open-source software.

Impact & stakeholders: Replicating findings of gender differences in personality and depression, anxiety, and stress will improve the knowledge base of clinicians who provide mental health treatment.



Background & Related Work

- Replication research is vital for improving our understanding of human behavior (Shrout & Rodgers, 2018, Youyou et al., 2023).
- Replication with slight methodological changes yields equally rigorous results, and advances the psychological sciences (Nosek & Errington, 2020; Shrout and Rodgers, 2018).
- Target studies for replication found:
 - Females tend to rate lower in emotional stability and higher in agreeableness compared to males (Weisberg et al., 2011);
 - Emotional stability is the best predictor of depression, stress, and anxiety (Uliaszek et al., 2010);
 - Confirmatory Factor Analysis (CFA) suggests DASS-21 is suitable for comparison of means between males and females (Gomez et al., 2014)



Methods & Evaluation

Approach: Analyses were conducted in R using R Studio, included R's built in independent samples t-test, and packages such as lavaan, and effectsize

Validation:

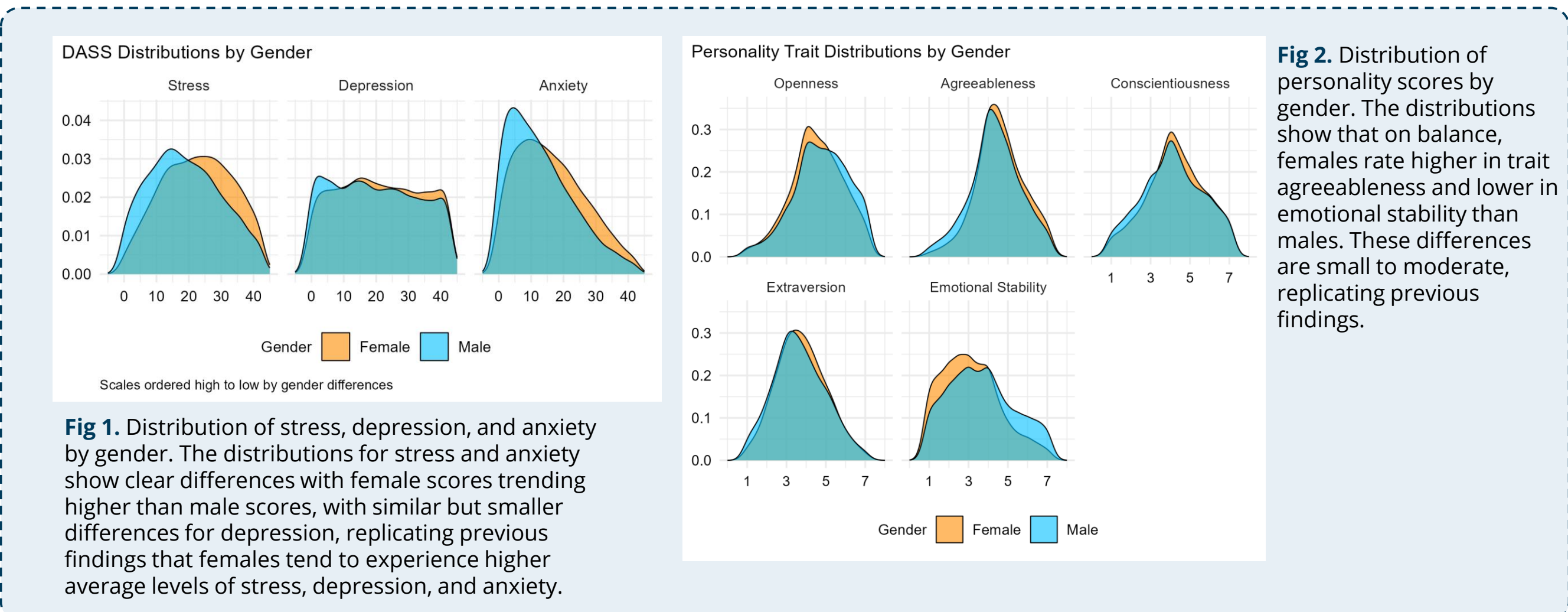
- Participants who failed an attention check were removed
- Given the large sample size:
 - Independent sample t-tests were reported by effect size
 - CFA was reported in terms of Comparative Fit Index (Furr, 2018)



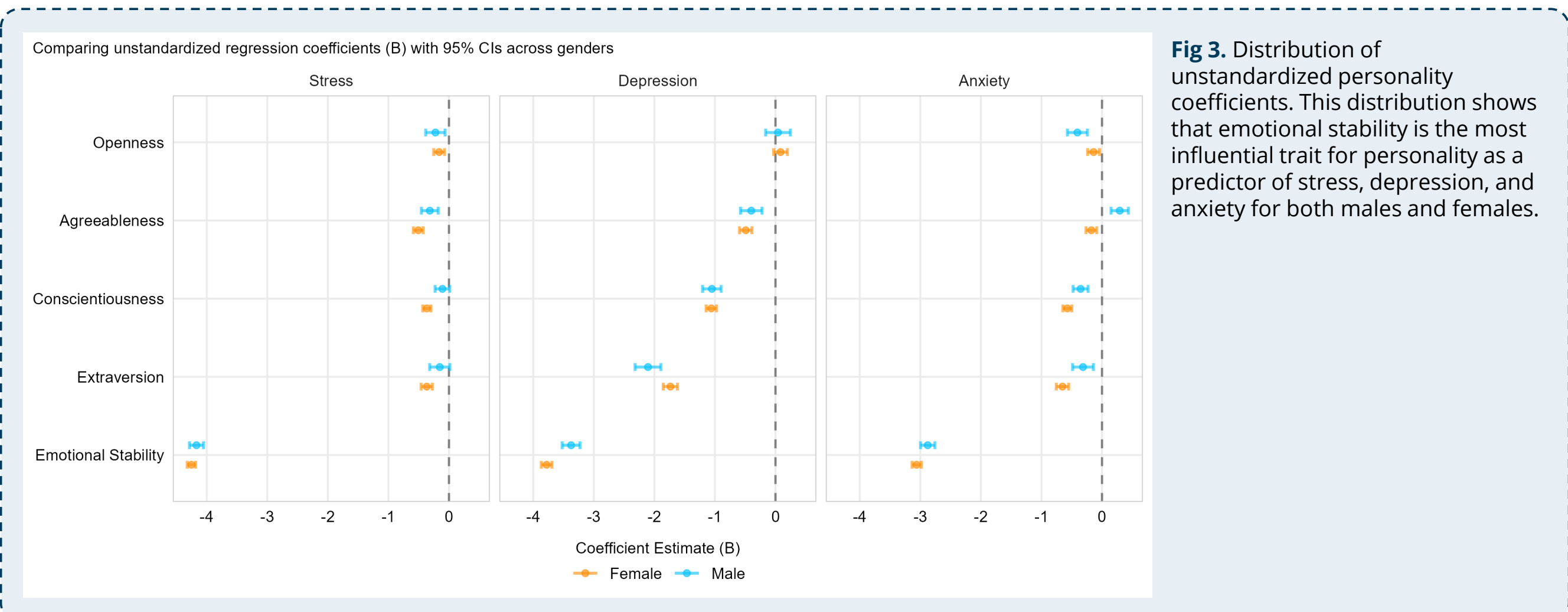
Results & Visualizations

- Males and females differ on trait agreeableness and emotional stability.
- Females, compared to males, reported higher levels of stress, depression, and anxiety.
- Emotional stability was the most influential trait for stress, depression, and anxiety.

DASS-21 & Personality Trait Distribution Plots



Personality Coefficient Plot for DASS-21 Indicators

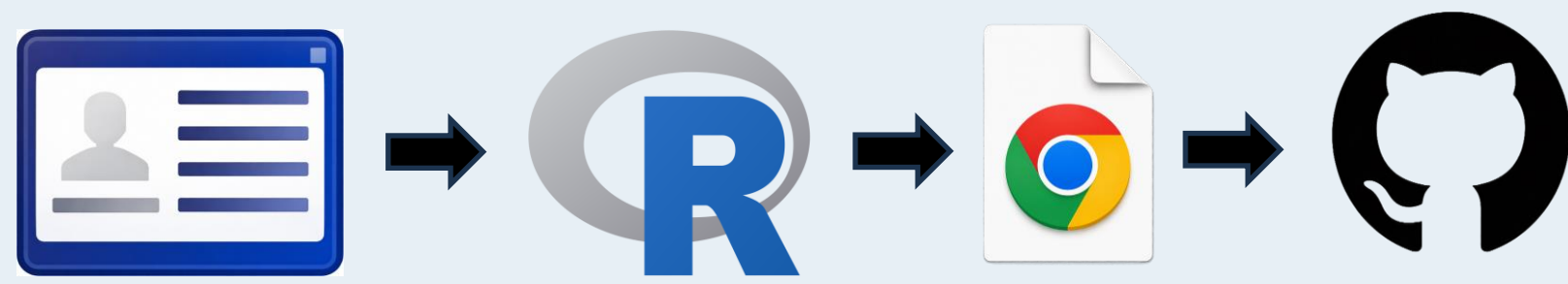


Data, Engineering & Ethics

Sources & scale: Open-Source Psychometric Project (2019). The data set contained 39,775 responses to the DASS-21 (Lovibond & Lovibond, 1995) and the Ten-Item Personality Inventory (Gosling et al., 2003).

Pipeline: CSV → cleaned in R → saved as R Markdown HTML → hosted on Git Hub

Ethics & responsible use: Data was collected and cleaned by Open-Source Psychometric Project for public use. There are no known ethical concerns.



Discussion, Limitations & Conclusions

Discussion: Results supported prior findings, adding to the “credibility revolution” described by Korbmacher et al. (2023). Therefore, this study adds to the growing body of replication studies advancing the cause of repairing public trust in the psychological sciences.

Limitations & future work: This survey was conducted online and may have been completed multiple times by the same participant. Additionally, populations with limited or no access to the internet were under sampled. Results from this study will be used to isolate important features from machine models that predict depression, stress, and anxiety in males and females.

Conclusions: This study showed open-source software can be used to conduct rigorous analysis of open-source data to replicate findings from previous research, thus opening the field of psychological science to a broad swath a researchers looking to contribute to the credibility revolution.



References

- Furr, M. (2018). Psychometrics: An introduction (3rd ed.). SAGE Publications Inc.
- Gomez, R., Summers, M., Summers, A., Wolf, A., & Summers, J. (2014). Depression Anxiety Stress Scales-21: Measurement and structural invariance across ratings of men and women. *Assessment*, 21(4), 418–426. <https://doi.org/10.1177/1073191113514106>
- Gosling, S. D., Rentfrow, P. J., & Swann, W. B. (2003). A very brief measure of the Big-Five personality domains. *Journal of Research in Personality*, 37(6), 504–528. [https://doi.org/10.1016/S0092-6566\(03\)00046-1](https://doi.org/10.1016/S0092-6566(03)00046-1)
- Korbmacher, M., Azevedo, F., Pennington, C. R., Hartmann, H., Pownall, M., Schmidt, K., Elsherif, M., Breznau, N., Robertson, O., Kalandadze, T., Yu, S., Baker, B. J., O'Mahony, A., Olsnes, J. Ø.-S., Shaw, J. J., Gjoneska, B., Yamada, Y., Röer, J. P., Murphy, J., ... Evans, T. (2023). The replication crisis has led to positive structural, procedural, and community changes. *Communications Psychology*, 1, 3. <https://doi.org/10.1038/s44271-023-00003-2>
- Lovibond, S. H., & Lovibond, P. F. (1995). Manual for the Depression Anxiety Stress Scales (2nd ed.). Psychology Foundation. <https://www2.psy.unsw.edu.au/dass/>
- Nosek, B. A., & Errington, T. M. (2020). What is replication? *PLoS Biology*, 18(3), e3000691. <https://doi.org/10.1371/journal.pbio.3000691>
- Shrout, P. E., & Rodgers, J. L. (2018). Psychology, science, and knowledge construction: Broadening perspectives from the replication crisis. *Annual Review of Psychology*, 69(Volume 69, 2018), 487–510. <https://doi.org/10.1146/annurev-psych-122216-011845>
- Uliaszek, A. A., Zinbarg, R. E., Mineka, S., Craske, M. G., Sutton, J. M., Griffith, J. W., Rose, R., Waters, A., & Hammen, C. (2010). The role of neuroticism and extraversion in the stress–anxiety and stress–depression relationships. *Anxiety, Stress, & Coping*, 23(4), 363–381. <https://doi.org/10.1080/10615800903377264>
- Weisberg, Y. J., DeYoung, C. G., & Hirsh, J. B. (2011). Gender differences in personality across the ten aspects of the Big Five. *Frontiers in Psychology*, 2. <https://doi.org/10.3389/fpsyg.2011.00178>
- Youyou, W., Yang, Y., & Uzzi, B. (2023). A discipline-wide investigation of the replicability of Psychology papers over the past two decades. *Proceedings of the National Academy of Sciences*, 120(6), e2208863120. <https://doi.org/10.1073/pnas.2208863120>



Reproducibility

Repository: [<https://github.com/your-org/your-project>]